ALIoT Smart Lab - Eight Upgrades Report

17 October 2026 (afternoon) · Portal **http://192.168.1.200** (screen sharing from the browser needs **https://192.168.1.200**) · branch `offline-router-door-photos`, not committed, not pushed

Status labels

- **VERIFIED**: built, tested and checked on the running portal.
- **PHYSICAL VALIDATION REQUIRED**: built and tested offline; still needs a real device or PC.
- **PARTIAL**: part is done, part is not.
- **BLOCKED**: needs a licence, IT approval or lab access.

Nothing in this round sent a command to a robot, door or PC. Robot control stays in DRY RUN.

1. The eight workstreams

Workstream	What was built	Tests / live result	Status
1 System health	Network checks (face server, Ollama, n8n, knowledge service, ESP32-CAM) now run in a background monitor: in parallel, every 10 s. Pages read its cache, so a request never waits on a probe. Each item shows ONLINE / DEGRADED / STALE / OFFLINE / UNKNOWN / SIMULATED, last OK, failing since, latency, open alerts and what to check first. A robot collector with no joint data shows as DEGRADED, and a camera service with no picture as DEGRADED. Data freshness and backup status are included. Shown on the System analytics page.	3 new tests. Live: `/system/health` median 37 ms, p95 63 ms. It reports the real state: 7 online, 3 degraded, 5 offline, 2 unknown.	**VERIFIED**
2 PC fleet and screen sharing	**Viewing a PC:** pick one monitor or all (the agent reports its monitors and re-reads them every frame), choose quality and frame rate (Low/Standard/High: 1/2/4 fps), full screen. The start and end of each viewing are audited. A "View all screens" wall and a "Using: app" line per PC. **Sharing to students:** pick the screen or window in the browser's own picker, check a local preview that nobody else sees, then Go live. Send to all lab PCs or selected PCs, or share from the master PC instead. The receiving PCs get a notice and open /share. Nothing is recorded. **Kept:** approved apps, lock, log off, restart, shut down.	17 equipment tests. Live (two real browsers): preview shown before going live, students saw the screen about 1.1 s after Go live, Stop ended it for them. The agent found this laptop's 2 monitors.	**PHYSICAL VALIDATION REQUIRED: no lab PC has the agent yet (needs IT approval). Pillow isn't installed here, so agent frame capture is untested. Remote keyboard/mouse is not built; Windows RDP shadowing with the user's consent is documented instead.**

3 Robot digital twins	UR5e Home to the pendant pose [-15.00, -126.00, 117.03, -80.00, 91.00, 95.27]°, confirmed from measured joints within 0.5°. Global STOP with latch and admin recovery. A collector reporting with no joint data shows as BLOCKED / DEGRADED, never READY. FANUC keeps its own adapter (monitoring only).	Robot control and robot tests pass. Live: UR5E_01 shows BLOCKED (no joint feedback) and mode is dry_run.	**PHYSICAL VALIDATION REQUIRED (UR5e). BLOCKED (FANUC: licensed remote interface unknown).**
4 Incidents	Every alert is an incident (INC-000123): acknowledge, investigate, assign an owner (staff only), notes, recovery actions, AI hypotheses (labelled), resolve (a reason is required), reopen. Overdue escalation flags (critical after 15 min, warning after 4 h). Related records from other systems in the same lab within ±10 min are shown **by time only**; a person is named only if the record itself names them. Everything is audited. New Incidents page.	4 new tests. Live: 68 open, 63 past their escalation time (never acknowledged).	**VERIFIED**
5 Anomalies and predictive maintenance	Three layers on real measurements: fixed limits, sensor-fault checks (impossible values, a frozen value, no data), and each node's own 7-day robust baseline. Each finding states the signal, expected range, observed value, time window, confidence and source, and can be raised as an incident once per day. Machine learning is not used and the page says why: one node with 6 days of data isn't enough.	4 new tests. Live: the environment node has been silent since 11 Oct. Temperature and humidity are within baseline (z 0.29 / -1.23). Real finding: the **flame detector was active in 17% of 4,095 readings**, so check its sensitivity.	**PARTIAL: vibration, motor current and energy aren't measured by any sensor, so they aren't analysed.**
6 Equipment lifecycle and inventory	Reports Centre tabs: inventory discrepancies (last staff count + scale changes vs the scale now, unattributed withdrawals; a count is the auditable correction) and equipment lifecycle (custodian, overdue check-outs, maintenance due, open fault reports, last movement). Existing gallery images and equipment lists are unchanged, and the Power Lab has no SCADA item.	2 new tests. Live: 26 assets, 1 flagged (PLC-001: 2 open fault reports). The bin can't be compared yet because no staff count exists.	**VERIFIED**

7 Security, backups, recovery	A sweep test calls all 258 API routes without signing in; only the explicit public list may answer. Students are refused on the new admin features. Sign-in throttling (already present) checked. Backup verified by a restore drill into a separate database: every table matches the live one. The new migration was tested upgrade → downgrade → upgrade on that copy before it was applied.	2 sweep tests pass. Automatic backups: last one OK (every 6 h).	**VERIFIED**
8 Thesis evidence	`py -I tools/evidence/run_evidence.py`: read-only checks of the live system. Writes a timestamped Markdown and JSON report with procedure, expected, observed, pass/fail, and evidence class (REAL_HARDWARE / LIVE_SYSTEM / OFFLINE / SIMULATION).	Latest run: 32 pass, 12 fail (devices off or at the university), 9 not available, 2 simulation. File: `docs/reports/evidence_20261017_124648.md`.	**VERIFIED (the software); the failures are honest hardware gaps**

2. Performance (live portal, this laptop over LAN, Chrome)

Measurement	Morning (before fixes)	Previous round	Now
Sign-in → app ready	3.6 s	0.95 s	0.83 s
Sidebar navigation, page shown (median)	-	80-130 ms	55-115 ms
Sidebar navigation, data loaded (median)	-	140-200 ms	139-217 ms
9 main API reads, 20 calls each, p95	-	-	34-66 ms (all under 70 ms)
`/system/health`	probes inline (up to 3.6 s)	-	p95 63 ms
Anomaly scan	-	0.87 s every load	0.45 s once a minute, 9 ms cached

Still slow-ish: the start page's data takes about 0.53 s (about 20 widget requests), and System analytics about 0.3 s. The /admin/equipment figure in the test is mostly the script scrolling to the link.

3. AI agent

Question type	Path	Time	Accuracy check
Lab data (free labs, offline devices, door history)	Straight from the database	30-100 ms	pass
Project / technical (quick)	Cited passages from project files	0.13-0.45 s (one cold run 2.6 s)	pass: right files cited
Project / technical (written answer, Project knowledge tab)	Local 1.5B model on the CPU	sources at 0.1-1.3 s; first word 8-17 s; full answer 10-19 s	After a doc wording fix it states the exact Home angles and the 0.5° rule, with valid citations.

The remaining delay is the CPU model reading its prompt before it writes. The GPU is kept for the cameras by your rule. A faster written answer would need the GPU or a larger machine.

4. Database, migration and backup

- Dump `backups/smartlab_before_incidents_20261017_152746.dump` (all 41 tables), restored into `smartlab_restore_drill`; table counts match the live database.
- Migration `a9d4e7c1b305` is additive (new columns and a new table, nothing dropped). On the drill copy: upgrade, downgrade and upgrade again all worked. On the live database: applied at the backend restart (306 resolved and 68 open incidents backfilled).
- The drill database is still there for reference. It's a copy, not production; delete it only if you want to.

5. Cameras and robots: real hardware vs simulation

- **Camera 2 (SECCAM_LAB02):** real picture, 7 fps, analysis 0.1 s old, no errors (REAL_HARDWARE, at home).
- **Camera 1 (SECCAM_LAB01, MC210):** service running, no picture (camera is at the university).
- **Tracking tests** (crossings, occlusion, several people): not done, because they need both cameras mounted at the lab.
- **UR5e and FANUC:** no live joints (robots at the university). The UR5e collector's data is now stale. Home and STOP are tested with mocks only, in DRY RUN.
- **Simulated robots** (UR5E_SIM, FANUC_SIM) are labelled SIMULATION and never counted as physical evidence.

6. Next physical actions (need IT, the lab or your sign-off)

1. **IT approval:** install `tools/pc_agent` (with Pillow) on one pilot PC and the admin PC. Steps are in the agent README; nothing inbound through the firewall.
2. **UR5e:** cable and IP, check "Home reached" using the pendant's own Home, clearance check, Remote mode. Then you sign off before `ROBOT_CONTROL=live`.
3. **FANUC:** find out which licensed remote option the controller has (socket messaging / PNS / UOP) before any control work.
4. **Cameras:** check camera 1's picture at the lab, mount camera 2, run the floor calibration, then crossing and occlusion tests. Re-run the evidence script afterwards.
5. **Flame detector:** check its sensitivity (17% activity rate). Restart the environment node, door controller and bins (offline since 11-12 Oct).
6. **Inventory:** do one staff count per bin, which gives the discrepancy view its baseline.

7. Files changed this round

- **Backend, new:** `services/health_monitor.py`, `services/anomalies.py`, `services/reports_center.py`, `api/routes/incidents.py`, `api/routes/reports_center.py`, migration `a9d4e7c1b305_incidents.py`.
- **Backend, changed:** `system.py`, `analytics.py`, `equipment.py`, `pc_control.py`, `admin.py` (resolve is audited), `models/tables.py`, `main.py`.
- **Backend tests:** `test_system_health.py`, `test_incidents.py`, `test_anomalies.py`, `test_reports_center.py`, `test_authz_sweep.py`, plus additions to `test_equipment.py`.
- **Frontend, new:** `SystemHealthBoard.tsx`, `AnomalyPanel.tsx`, `ReportsCenter.tsx`, `pages/Incidents.tsx`, `pages/Share.tsx`, `ScreenShare.tsx`, `lib/screenShare.ts`.
- **Frontend, changed:** `PcControls.tsx`, `NetworkedEquipment.tsx`, `Layout.tsx`, `App.tsx`, `lib/api.ts`.
- **Tools and docs:** `tools/pc_agent/pc_agent.py` (version 0.5) and its README, `tools/evidence/run_evidence.py`, `docs/ROBOT_CONTROL.md`, `docs/PROGRESS_CHECKPOINT.md`.

Tests: backend 492 passed; robot monitor 37, knowledge 40, vision 163; TypeScript check and production build OK.